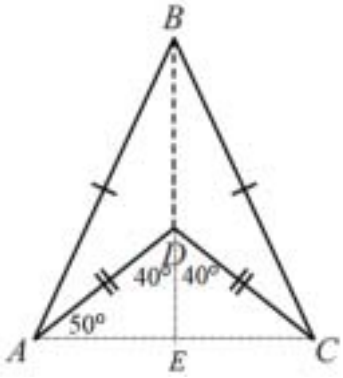
Question	Answer	Marks	Partial Marks
6(b)	203 or 202.6…	**3**	**B2** for [angle at W =] 22.6... or for [angle at V =] 67.4 or 67.38… or **M1** for tan = $\frac{5}{12}$ or $\frac{12}{5}$ oe
6(c)	25.2 or 25.20 to 25.21[0]  	**5**	**B4** for [BC or AB =] 7.6[0] or 7.604 to 7.605 OR **M3** for a complete explicit method leading to AB or BC, e.g. $\frac{5\sin 140}{\sin 25}$ OR **M2** for a complete implicit method leading to AB or BC, e.g. $\frac{\sin 25}{5} = \frac{\sin 140}{BC \text{ or } AB}$ oe and **M1** (dep on AB from trig) for 2 × *their* AB + 10 OR **B1** for any relevant angle E.g. $\angle BDA$ or $\angle BDC$ = 140, $\angle DAE$ or $\angle DCE$ = 50 or $\angle ADE$ or $\angle CDE$ = 40 or $\angle ADC$ = 80
6(d)	79.5 or 79.6 or 79.54 to 79.55...	**5**	**B2** for [PR^2 =] 245 or 245.1 to 245.2 or [PR =] 15.65 to 15.66 or 15.7 or **M1** for [PR^2 =] $11^2 + 8^2 - 2 \times 11 \times 8 \times \cos 110$ **M2** for [cosPQR =] $\frac{10^2 + 14^2 - (\textit{their } PR)^2}{2 \times 10 \times 14}$ oe or **M1** for (*their* PR)^2 = $10^2 + 14^2 - 2 \times 10 \times 14\cos PQR$ oe
7(a)(i)	40	**2**	**M1** for $\frac{50}{75}$ [× 60] oe
7(a)(ii)	36 nfww	**3**	**M2** for $\frac{47 - 0.5}{75 \text{ to } 80}$ [× 60] or $\frac{46 \text{ to } 47}{75 + 2.5}$ [× 60] or **M1** for 47+0.5 or 47 – 0.5 or 75 + 2.5 or 75– 2.5

Question	Answer	Marks	Partial Marks
7(b)	107 or 107.2...	**6**	**M5** for [speed =] $\frac{240}{\left(2\times\frac{260}{7}+60\right)}\times 60$ oe OR **B5** for [total time =] 134 or 134.2 to 134.3 or 2.24 or 2.238... or **B4** for (t =) 37.1 or 37.14... OR **M2** for $\frac{t}{60}\times 100+\frac{t+60}{60}\times 110=240$ oe or **M1** for $\frac{t}{60}\times 100$ or $\frac{t+60}{60}\times 110$ oe **M1** for correct equation of form $at = b$ from *their* equation containing two terms in t and involving the speeds. **M1** for $\frac{240}{2\times \textit{their}\, t+60}$ [× 60]
8(a)(i)	37.3 or 37.26 to 37.27	**5**	**M2** for $\pi\times 0.35\times\sqrt{0.35^2+1.5^2}$ oe or **M1** for $0.35^2+1.5^2$ or better **M1** for $\pi\times 0.35^2$ **M1** for $2\times\pi\times 0.35\times 16$
8(a)(ii)	6.35 or 6.349 to 6.351	**3**	**M1** for $\pi\times 0.35^2\times 16$ **M1** for $\frac{1}{3}\times\pi\times 0.35^2\times 1.5$
8(a)(iii)	22.2 or 22.3 or 22.24 to 22.26	**3**	**M2** for 17.5 × 3.5 × 1.4 – 10 × *their***(a)(ii)** or **M1** for 17.5 × 3.5 × 1.4